

Answer Key — Biology Grade 11

Sub-Strand 2.1: Reproduction in Plants • Lesson 5: Inside the Fruit: Placentation as Evidence of Post-Fertilisation Development

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
B	D	A	D	C	B	C

USE AFTER: PREDICT PHASE — Predict: learners predict which part of the flower becomes the fruit

Q1. After a flower has been fertilised, which part of the flower becomes the fruit and which part becomes the seed?

- A. The ovule becomes the fruit and the ovary becomes the seed
- B. The ovary becomes the fruit and the ovule becomes the seed**
- C. The ovary becomes the seed and the stigma becomes the fruit
- D. The petals become the fruit and the ovule becomes the seed

Correct answer: B. After fertilisation the ovules inside the ovary develop into seeds, and the ovary wall develops into the fruit. The other options swap the parts or use parts that wither.

USE AFTER: OBSERVE PHASE — Observe: learners locate the placenta and seed attachment point and record where seeds are attached in a table

Q2. What is placentation in a fruit?

- A. The way a fruit changes colour and softens as it ripens
- B. The way pollen lands on the stigma of a flower
- C. The number of seeds that a fruit can produce
- D. The way ovules or seeds are attached to the placenta inside the ovary or fruit**

Correct answer: D. Placentation is the arrangement of ovules (later seeds) on the placenta inside the ovary. It is not pollination, ripening or seed number, which are common confusions.

USE AFTER: OBSERVE PHASE — Observe: sectioning tomato and orange (transverse section)

Q3. A learner cuts a tomato across (transverse section). She sees 2 to 3 chambers, and the seeds are attached to a central column where the walls between the chambers meet. Which type of placentation does the tomato show?

- A. Axile**
- B. Free-central
- C. Parietal
- D. Marginal

Correct answer: A. Seeds attached to a central axis in an ovary divided into several chambers (locules) is axile placentation, as in tomato and orange. Free-central has a central column without partitions.

USE AFTER: OBSERVE PHASE — Observe: sectioning passion fruit and cucumber (transverse section)

Q4. A passion fruit is cut across. It has one single chamber and the seeds are attached to the inner wall of the fruit. Which type of placentation is this?

- A. Axile, because the seeds are attached to a central axis
- B. Marginal, because the seeds are attached along one seam
- C. Free-central, because there is a column in the middle
- D. Parietal, because the seeds are attached to the wall of a single chamber**

Correct answer: D. Seeds on the wall of a single-chambered ovary show parietal placentation, as in passion fruit and cucumber. Marginal placentation has seeds along only one edge or seam, as in a bean pod.

USE AFTER: EXPLAIN PHASE — Explain: learners explain why a lone avocado or pawpaw with flowers but no pollination from another tree gives no seeds and no fruit

Q5. A lone pawpaw tree in a compound is covered in flowers but bears no fruit, and there is no other pawpaw tree nearby. Which explanation is best?

- A. The ovules turned into fruit without the ovary, so no seeds could form
- B. The petals normally become the fruit, and they fell off the tree too early
- C. The flowers were not pollinated and fertilised, so the ovules did not become seeds and the ovary did not become a fruit**
- D. The tree has the wrong placentation, so the flowers cannot produce fruit

Correct answer: C. Fruit and seeds form only after fertilisation. With no pollen from a partner tree there is no fertilisation, so the ovules and ovary do not develop. Placentation describes seed arrangement, not whether fruit forms.

USE AFTER: EXPLAIN PHASE — Explain: teacher corrects the misconception that 'the fruit is only the sweet part' by asking whether a bean pod is a fruit

Q6. A learner says, "A bean pod is not a fruit because it is dry and not sweet like a mango." Which response is correct?

- A. The learner is wrong, because a bean pod is a seed that has grown larger
- B. The learner is wrong, because the bean pod is a dry fruit: it is the developed ovary and it contains the seeds**
- C. The learner is right, because the pod is a leaf that protects the seeds
- D. The learner is right, because only sweet and juicy structures are fruits

Correct answer: B. A fruit is a developed ovary containing seeds, whether it is sweet or dry. The bean pod (a legume) is a dry fruit with seeds attached along its seam.

USE AFTER: MODEL BUILDING PHASE — Model: learners keep the parallel branch of vegetative propagation (cassava, sugarcane, sweet potato cuttings) on the class model

Q7. A farmer plants a piece of cassava stem and it grows into a new plant. Which statement correctly describes how this new plant formed?

- A. It formed after the cutting was pollinated by pollen from another cassava plant
- B. It formed because the cutting first made seeds inside a fruit
- C. It formed without flowers, fertilisation or seeds, because the cutting grows directly into a new plant**
- D. It formed after double fertilisation inside the cutting

Correct answer: C. Vegetative propagation by cuttings bypasses flowers, pollination, fertilisation and seeds. This is the parallel branch of the class model, separate from the flower to seed to fruit route.